# FC01 compact two-nozzle filling cell

## Siemens-authoritative controls and bounded NVIDIA quality architecture

**FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION**

**CONCEPTUAL SAFETY ARCHITECTURE - REQUIRES PROJECT-SPECIFIC RISK ASSESSMENT, DESIGN, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY-SAFETY ENGINEER. NO PERFORMANCE LEVEL, SIL, CATEGORY, CE CONFORMITY OR REGULATORY COMPLIANCE IS CLAIMED.**

| Release status | What is implemented | What remains blocked |
|---|---|---|
| NVIDIA IMPLEMENTATION-READINESS RELEASE CANDIDATE | Revision-F canonical model; Git-object byte authority; 47-node immutable PLC/AI contract; production-shaped edge and dataset tools; 32 process scenarios plus 28 structured behavioral vision fault injections; inherited native CAD/QET evidence. | TIA/WinCC compile/archive; Startdrive; PLCSIM; final site electrical inputs; real dataset/model/target runtime; physical FAT/SAT and qualified safety work. |

This report is an evidence index. It is not proof of construction readiness, functional safety, native compile, FAT, SAT, electrical test, physical commissioning or AI performance.

# Integrated architecture and deterministic ownership

The PLC owns all machine sequence, interlocks, timeouts, product disposition and transfer permission. THE NVIDIA VISION SUBSYSTEM IS NON-SAFETY-RELATED AND MUST NOT BE USED AS THE SOLE MEANS OF PERSONNEL PROTECTION, SAFE STOP, GUARD MONITORING OR HAZARDOUS-MOTION CONTROL.

| Layer | Owner | Deterministic responsibility | Failure behavior |
|---|---|---|---|
| Field / power | Electrical | Identified device, cable/core, terminal and I/O channel | Fail-closed valves; external safety removes hazardous energy |
| Equipment control | S7-1500 FB instances | Gate, clamp, two fill channels, conveyor/pump VFD, capper | Module timeout/contradiction fault; outputs safe |
| Machine coordinator | S7-1500 | Legal states, accepted command edges, no automatic restart | HOLDING, CONTROLLED_STOPPING or FAULTED |
| Operator interface | MTP700 Unified | Role-checked handshake; feedback and disabled reason | No direct physical-output writes |
| Quality vision | NVIDIA edge service | Securely acquire request and publish one correlated immutable result | Malformed/stale/low-confidence/fault -> quality hold |
| Regression | Python + asyncua | Sequence fault injection and encrypted real server/client transport | Design evidence only; never PLCSIM or production-endpoint proof |

Machine states: UNINITIALIZED, INITIALIZING, STOPPED, READY, AUTOMATIC, MANUAL_SETUP, HOLDING, CONTROLLED_STOPPING, FAULTED and RECOVERY_RESET. Reset returns to STOPPED and never initiates motion.

# Selected Siemens and edge hardware baseline

| Function | Selected item | Order number / status | Verification boundary |
|---|---|---|---|
| PLC | SIMATIC S7-1500 CPU 1511-1 PN | 6ES7511-1AL03-0AB0 | Current official lifecycle checked; V20 device catalog/compile open |
| Pulse counting | TM Count 2x24V | 6ES7550-1AA01-0AB0 | Two independent 24 V pulse channels; TIA configuration open |
| DI / DO / AI | DI 32 HF; DQ 32 HF; AI 8xU/I HF | 6ES7521-1BL00-0AB0; 6ES7522-1BL01-0AB0; 6ES7531-7NF00-0AB0 | Front connectors, wiring and catalog compatibility open |
| HMI | MTP700 Unified Comfort | 6AV2128-3GB06-0AX1 | WinCC V20 compile/licence open |
| Drives | Two SINAMICS G120C PN 0.75 kW | 6SL3210-1KE12-3UF2 | Ratings provisional; Startdrive absent |
| Network | SCALANCE XC208 managed plus S615 firewall | 6GK5208-0BA00-2AC2; 6GK5615-0AA01-2AA2 | VLAN/firewall architecture selected; native policy and site approval open |
| Vision | Jetson Orin NX 16GB on industrial carrier | Carrier/integration SKU TBD | Carrier, thermals, EMC, camera and real dataset open |

Authoritative engineering target: TIA Portal V20, executable/product version 2000.0.9501.1. Installed STEP 7 and WinCC V20 components do not prove usable licence entitlement. Current identity is not authorized for TIA Openness. Startdrive and PLCSIM are absent.

# PLC-AI v3 acceptance contract

Transport selection: encrypted OPC UA across a routed quality zone. The PLC publishes one immutable request and holds its request until coherent capture acknowledgement or a terminal result. Results are accepted only for the current session/ID with fresh heartbeat, healthy service/camera/model, approved model/dataset/calibration identities, allowed disposition and reasons, sufficient confidence, valid timing and a non-contradictory payload.

| PLC-owned nodes (13) | NVIDIA-owned nodes (34) | Acceptance rule |
|---|---|---|
| Enable, trigger, inspection/session identity, recipe/count/fill target, PLC heartbeat, exact result ACK, maintenance mode and expected dataset/calibration IDs | Ready/busy/capture/state/health; immutable quality payload; disposition/reasons/confidence; model/dataset/calibration identities; diagnostic timestamps/timing; edge heartbeat/diagnostics/queue | Payload publishes before RESULT_VALID and remains immutable until exact ACK. PLC-derived monotonic result age is authoritative; edge UTC is diagnostic. Any stale, malformed, unhealthy, uncertain or contradictory condition enters HOLD/FAULT. |

The adapter requires Basic256Sha256 SignAndEncrypt, X.509 application/user identity, controlled trust/CRL stores and an exact 47-node typed map. Local certificate-backed asyncua tests use a synthetic endpoint. No model is delivered. No training, ONNX export, TensorRT engine, DeepStream execution, accuracy, confusion matrix, false-accept/reject analysis or production latency is claimed. The RTX 5060 Laptop GPU does not substitute for the absent CUDA/TAO/DeepStream target stack or representative dataset.

# NVIDIA implementation-readiness engineering

| Work package | Revision-F controlled result | Open acceptance boundary |
|---|---|---|
| Acquisition/runtime | Exactly-once recorded-image source, bounded preprocessing, production-prohibited mock backend and hash-gated ONNX adapter boundary | Physical camera/lens/light, actual tensor contract and target runtime |
| Dataset/evaluation | Versioned annotation/manifest schemas, file hashes, lot/session split controls, duplicate/leakage checks and FAR/FRR evaluation math | Representative independently labeled multi-lot data, approved thresholds and model-change approval |
| Platform policy | DeepStream 9.1 and TAO 7.0.1 tracked; generic TensorRT 11.1 is not treated as a Jetson compatibility lock | Selected JetPack/carrier image, CUDA/TensorRT/DeepStream/TAO execution and deployment validation |
| Electrical delta | Separate provisional 24 V branches: 60 W edge, 12 W camera, 20 W lighting and 8 W service/network; X200 and switch-port reservations | Selected devices, inrush, protection, cable, voltage-drop, thermal, EMC and native schematic/CAD incorporation |
| Service lifecycle | Schema-controlled configuration, structured diagnostics, durable event-store boundary, health/metrics, rollback and troubleshooting procedures | Site PKI/time/retention/privacy policy, production endpoint and operational acceptance |

**Synthetic fixtures and local timing harnesses prove software plumbing only. They cannot authorize production READY, establish optical feasibility, or support any accuracy, false-accept, false-reject, throughput or latency claim.**

# Validation evidence and release blockers

| Gate | Evidence | Status |
|---|---|---|
| Canonical consistency | 551 engineering-validator checks plus 96 dedicated Revision-F checks cover generator parity, schedules, ownership, edge behavioral fault injection, safety boundaries and prohibited artifacts. The standalone electrical delta is not incorporated into every authority. | PARTIAL |
| Deterministic regression | 99 Siemens/simulator/source/native-contract + 86 edge-service + 17 interface-harness tests; 32 process and 28 structured behavioral fault-injection cases | SOURCE/LOCAL PASS - not PLCSIM/HIL |
| Workbook | 26 summary/schedule sheets; deterministic normalized build, formula-error scan and all-sheet rendered review | PASS after final freeze |
| QElectroTech | Corrected 26-folio QET BCA022BE0B9F... reopened at the exact hash and exported natively to 26 pages. All pages reviewed; folio 25 has one clipped in-body safety sentence and automatic cross-reference resolution is unsupported | NATIVE REOPEN/EXPORT PASS; OVERALL PARTIAL |
| Panel CAD | FreeCADCmd reopened 165 objects / 148 controlled solids; STEP solid, IGES bounded-face and DXF/hole reimports verified | PASS |
| TIA / WinCC / drives | V20 installed; no native project, compile, cross-reference, archive restore or Startdrive evidence | BLOCKED |
| PLCSIM | Not installed | BLOCKED |
| NVIDIA software | 86/86 edge tests including 10 certificate-backed local asyncua cases; 47-node map, terminal-before-valid ordering, durable-audit failure interlock and 28-case behavioral injection pass | PARTIAL - production endpoint/native binding open |
| NVIDIA model/runtime | Production PLC/PKI, representative dataset, trained model and CUDA/TensorRT/DeepStream target stack absent | MODEL/RUNTIME BLOCKED |

Gate totals: 3 PASS, 7 PARTIAL, 11 BLOCKED, 1 OPEN. Release use: controlled design review and continuation in qualified native tools only. Read the manifest, acceptance-gate report and open-input register before work.